Linux Fundamentals – Lesson 1

Type	Draft
Status	Not started
Project	My DevOps Journey 🚀

1. What is Linux?

First, look at your laptop.

Assume your laptop is Windowa 11

Your laptop has Windows 11, right?

Windows helps you:

- Open applications.
- Save files.
- Browse the internet.
- Connect to Wi-Fi.
- Run programs.

Linux does the same work as Windows.

Linux is also an Operating System.

Definition :

Linux is an open-source operating system that helps us run applications and manage computer hardware and files.

Open-source means its source code is available for people to study, use and modify.

2. Why do we use Linux?

Linux is commonly used by companies to run their websites and applications.

For example:

Imagine you created a food delivery application.

Now you want people to use your application through the internet.

You need a computer called a server to run your application.

Many companies use Linux on their servers.

Linux helps us:

1. Run applications.
2. Store files.
3. Manage servers.
4. Connect to networks.
5. Run commands.

Remember: Linux is important in DevOps because many company servers use Linux.

3. What is an Operating System?

An Operating System (OS) is software that helps the user communicate with the computer.

Examples:

- Windows
- Linux
- macOS
- Android
- iOS

Easy example:

When you open Chrome, Windows helps Chrome run on your laptop.

Similarly, Linux helps applications run on a Linux computer.

4. What is a Linux Distribution?

Linux has different versions. These versions are called Distributions (Distros).

Some examples:

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Ubuntu

Very good for beginners.

Software We Use At NixNexus - NixNexus - Your Guide to everything FOSS

Debian

Known for stability.

RHEL, Red Hat Enterprise Linux | Piros

Red Hat Linux

Used by many companies.

We will use Ubuntu to practise Linux.

5. What is a Terminal?

A terminal is a place where we type commands to give instructions to the computer.

For example:

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pwd
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This command tells us where we are currently working.

There are two ways to use a computer:

- GUI – Click on icons, folders and buttons.
- CLI – Type commands to perform work.

6. Important words to remember

Word	Simple meaning
Linux	An operating system
OS	Software that runs a computer
Server	A computer that provides services to other computers
Terminal	Place to type commands
Command	An instruction given to a computer
GUI	Working using icons and buttons
CLI	Working using typed commands
Distribution	A version of Linux